Letters

Improved Safe Operating Area Modeling of DC-Link Capacitor Considering Transient Electro-Thermal Procedure

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***Abstract*—This article proposes an improved safe operating area (SOA) modeling of DC-link film capacitor considering the transient electro-thermal process. An in-depth analysis and evidence of the failure mechanism are provided accounting for the impact of leakage current and corresponding electro-thermal process. Then, an electro-thermal modeling of DC-link capacitor is proposed under severe operation conditions which considers the strong coupling between leakage current and temperature. Finally, the improved SOA model is derived and verified through *proof-of-concept* experiments.**

***Index Terms*—DC-link, failure machine, film capacitor, safe operating area (SOA), thermal model.**

I. Introduction

DC-LINK capacitors are one of the key components in power electronics and are widely used in traction drives, wind power, and photovoltaic inverters [1], [2], [3]. Particularly in large-scale PV medium-voltage DC collection systems, these film capacitors are deployed as critical energy-buffering elements which are continuously subjected to sever electrical and thermal stress. Consequently, the failure of the capacitor can lead to the high hardware and maintenance costs. Hence, its necessary to obtain their safe operating area (SOA).

There are two main failure mechanisms of film capacitors: one is catastrophic failure induced by single-event stresses, such as overvoltage and overheating; the other is wear-out failure including design defects, degradation, etc. [4]. Most studies focus on the wear-out failure: aluminum electrolytic capacitors degrade because of chemical and electrochemical reactions on electrode surfaces [5]; the migration of oxygen vacancies at the cathode results in the failure of multilayer ceramic capacitors [6]; the metallized polypropylene film capacitors fail mainly due to dielectric loss [7].

The catastrophic failure during operation mainly results from overvoltage and overcurrent, and can be categorized as open- and short-circuit failure. The open-circuit failure is mainly caused by self-healing dielectric breakdown, while the short-circuit failure is caused by dielectric breakdown or cracking. In this letter, we focus on the open-circuit failure of film capacitors. Typically, the film capacitor datasheet from the manufacturer gives the limits of voltage, current, and temperature. However, those conditions are tested under individual operating points, which only reflects single scenarios and fails to cover different real-world operation conditions [8]. Moreover, these test conditions are ideal, and do not consider degradation and tolerance. Therefore, an accurate operating limit is essential to guide the selection of proper film capacitors in severe DC-link applications. Our previous work [8] proposed the SOA concept for film capacitors defined by voltage and current ripple. It mainly focuses on the impact of ripple current on a timescale of tens of hours in steady-state operation. Hence, the original SOA ignores the transient thermal process on minutes to hours scale in the small-time span. Our recent experiments find that there are still cases where the original SOA cannot guarantee the safe operation of the capacitor, especially for those intense transient electro-thermal processes, which will be discussed below.

The open-circuit failure of the capacitor is mainly due to internal aging and extrinsic electrothermal reactions, while capacitor voltage stress, ripple current stress, and ambient temperature are all critical failure stressors. In recent years, most research have focused on the long-term aging process of the capacitor, neglecting short-timescale failure quantification closely related to thermal transient. In [9], the transient thermal model of aluminum electrolytic capacitors is proposed, containing the thermal to core resistance, the core to air resistance, and the core thermal capacitance. Rambetius [10] concluded that DC-link capacitor requires a two-thermal-node model with thermal capacitors due to their two distinct thermal time constants. These works mainly focus on the thermal domain, however, the electric domain is

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TABLE I
SPESIFICATIONS OF CUSTOMIZED CAPACITOR

C_r[uF]	$V_{C,max}$[V]	$I_{rp,max}$[A]	$T_{a,op}$[°C]	R_s[mΩ]	m[kg]	tanδ
645	1875	80	−40–85	0.06	1.9	0.0002

C_r: rated capacitance; $V_{C,max}$: maximum dc voltage; $I_{rp,max}$: maximum ripple current; $T_{a,op}$: ambient temperature for normal operation; R_s: sum of all Ohmic resistances occurring inside the capacitor; m: weight; $tan\delta$: dielectric dissipation factor in rated frequency

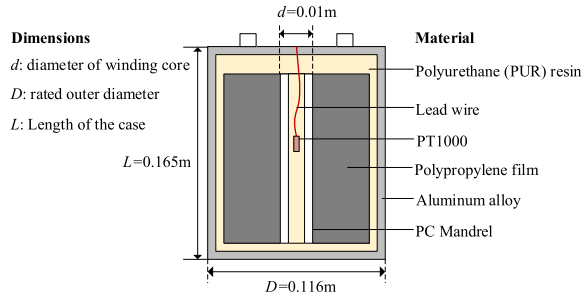


Fig. 1. Structure of the capacitor with thermal sensor.

TABLE II
TEST CONDITION OF EACH CAPACITOR

Test capacitor	DC voltage	AC ripple (50 Hz)	Oven temperature
No.1&2	1600 V	0	95°C
No.3&4	1600 V	0	85°C
No.5&6	1400 V	40 A	100°C

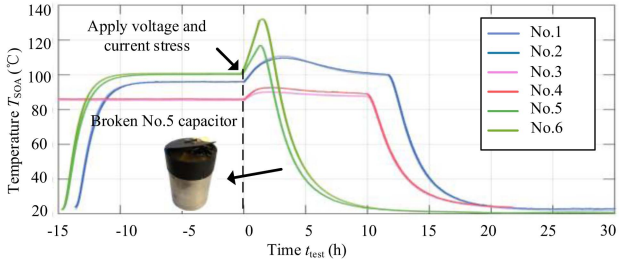


Fig. 2. Core temperature recording of each capacitor.

also required to determine the precise transient electro-thermal process and related SOA.

To address these issues, this letter investigates the improved SOA of the DC-link capacitors considering transient electro-thermal process and makes two contributions.

1) An in-depth analysis and evidence of failure mechanism of capacitors are provided based on different phases in the transient thermal and electrical processes.
2) An electro-thermal modeling of DC-link capacitor is proposed under severe operation conditions which considers the strong coupling between leakage current and temperature. The SOA is then determined combining the thermal model and thermal-related failure mechanism in small time scale.

The rest of this letter is organized as follows. Section II introduces the failure mechanism of capacitors considering the positive temperature feedback process. In Section III, the transient thermal model is constructed to obtain SOA, and it is verified in a case study. Finally, Section IV concludes this article.

II. DC CAPACITOR TRANSIENT FAILURE MECHANISM

DC capacitors are tested in this section, and the transient failure mechanisms under DC/AC ripple are analyzed in detail.

A. Test Procedure

The parameters of the customized capacitors E50.R16-654NT9 are shown in Table I. Thermal sensors PT1000 with a measurement range from −40 °C to 200 °C are pre-embedded inside the capacitor to measure the core temperature. Fig. 1 presents the detailed structure and material of capacitor including the position of PT1000. This modification introduces a negligible decrease in equivalent thermal resistance, leaves the dominant heat dissipation path unchanged as verified by our simulation, and does not affect either the leakage current or the dielectric integrity.

Tests on six capacitors under different electrothermal stresses were conducted in advance, with core temperatures recorded. Table II shows the test conditions and the test procedure is:

1) Discharge all capacitors using OEATE PR250 Series Thick Film Power Resistor with STAUBLI SiliStrom/SILI S Series Silicone Insulated Multistrand Wires.
2) Initialize the Keysight E4980A Impedance Analyzer. Use 16089B test fixture to perform open-circuit and short-circuit calibration.
3) Place 2 capacitors into VT4002 Chamber. Connect them in parallel to the SHEKONIC JZS-35 Test Power Supply; connect the HIOKI Power Analyzer PW6001 to the circuit to measure voltage and current; connect the PT1000 sensors to the Keysight Temperature Data Acquisition Unit.
4) Set target temperature, 0% humidity, and over-temperature protection at target temperature + 30 °C. Allow the system to stabilize for at least 6 h until thermal steady-state.
5) Apply electrical stress and monitor the temperature.
6) Immediately turn OFF the DC-power supply and chamber heating when the first capacitor fails (if no failure occurs after 12 h of continuous voltage application and the temperature reaches constant, stop the test).
7) Keep the chamber door closed and allow natural cooling for at least 6 h. Fully discharge all capacitors again before opening the chamber door.
8) Complete the test record by documenting the full voltage, current, temperature curves, and photographs.

Fig. 2 shows the core temperature of each capacitor. At 0 h, the voltage and current stresses are applied. The core temperature first increases to a peak and then decreases to a constant value. For no. 5 which failed, the temperature reaches a maximum of 131.7 °C when the electrical stress is removed manually. It is a thermal runaway and the applied stress is outside SOA. Hence, the transient thermal process can lead to the overheating and failure of the capacitor, which shall be considered when determining SOA.

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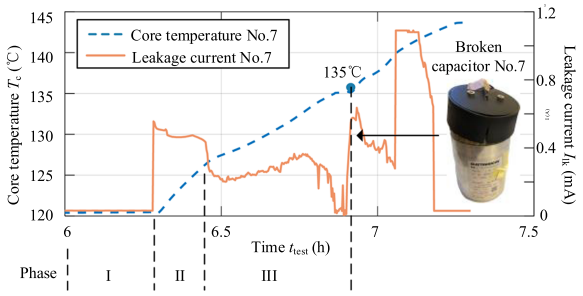


Fig. 3. Temperature and leakage current of capacitor no. 7.

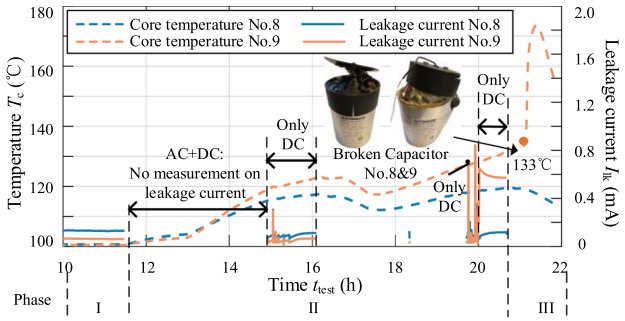


Fig. 4. Temperature and leakage current of two capacitors no.8 and 9.

B. DC Failure Mechanism

Capacitor 7 is tested under thermal and DC voltage stresses. Fig. 3 shows the measured leakage current and temperature, including three phases. In phase I, the capacitor is heated to 120 °C with no electrical stress applied. Phase II begins when 1500 V DC is applied. The leakage current and core temperature start to increase. In phase III, the temperature increases with a different slope. When the core temperature reaches 135 °C, it enters into a positive feedback loop due to the temperature increase caused by leakage current. At the end of phase III, the leakage current increase by about 50% again, and the capacitor fails with an open circuit.

In the failure process phase III, the leakage current leads to an increase in core temperature and vice versa. There is a critical positive temperature feedback point between phases II and III that can drive the capacitor into the thermal runaway stage. Therefore, it is essential to determine this positive feedback point to avoid entering phase III.

C. DC + AC Failure Mechanism

Capacitors 8 and 9 are tested with thermal, DC voltage, and AC current ripple stresses, as shown in Fig. 4. In phase I, the chamber heats capacitors to 100 °C. In phase II, 1400 V DC and 40 A AC current ripple are applied. When the temperature of Capacitor 9 approaches 120 °C, the capacitor is regarded as dangerous and close to the positive feedback stage. Hence, the AC ripple is stopped, leaving only DC excitation to measure and record the leakage current. If DC excitation fails to drive the capacitor into the critical positive feedback point, the AC current ripple is added again to increase the temperature further. Each time the temperature increases by 1 °C, the AC is shut down

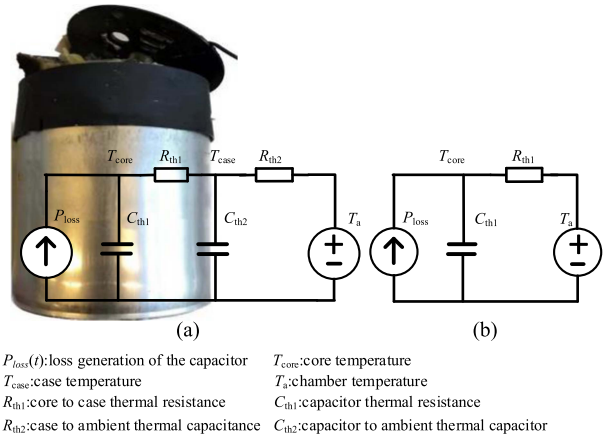


Fig. 5. Transient thermal model of capacitor. (a) Full circuit model. (b) Simple circuit model which only considers the capacitor body.

and the DC leakage current is recorded. This action repeats until the end of phase II. At the start of phase III, both AC and DC remain applied and the core temperature reaches 133 °C. Then, the temperature increases with much faster speed. Even when the AC ripple is removed and only DC remains, the temperature still increases quickly until 177 °C. Hence, phase III is the positive temperature feedback stage driving the capacitor to fail, and 133 °C is the positive feedback point which is similar to the DC failure case.

Therefore, the temperature is the critical indicator which leads to the catastrophic failure of the capacitor instead of the voltage and current stresses. The strong positive temperature feedback plays the dominant role leading to the catastrophic failure due to over temperature.

In conclusion, for both DC and DC + AC catastrophic failure, the transient temperature procedure is important to determine whether the capacitor can reach the positive feedback point. Moreover, even if the capacitor temperature with electrical stress is below the positive feedback point in steady-state, it can still fail in the transient temperature process.

III. Improved DC Capacitor SOA Modeling

A. Transient Thermal Modeling

The heat generated in the core transfers through the capacitor body to the ambient. Thus, a second-order model considering the thermal capacitance of the ambient and capacitor body is built in Fig. 5(a). The ambient temperature is precisely controlled by the oven, therefore the ambient to case thermal resistance and capacitance are neglected, as is simplified in Fig. 5(b). The SOA is obtained from the capacitor loss P_{loss}

$$P_{\text{loss}}(t) = \frac{T_{\text{core}} - T_a}{R_{\text{th1}}} + C_{\text{th1}} \frac{dT_{\text{core}}}{dt} \tag{1}$$

$$P_{\text{loss}}(t) = u_{\text{cap}} \times i_{\text{cap}} = i_{lk} \times V_{\text{DC}} + i_C^2 \times R_{\text{ESR}} \tag{2}$$

where u_{cap} and i_{cap} are the capacitor voltage and current, i_{lk} is the leakage current, V_{DC} is the DC voltage, i_{C} is the ripple current rms value, R_{ESR} is the equivalent series resistance which

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TABLE III
THERMAL CAPACITANCE AND RESISTANCE

V_{DC}(V)	T_a(℃)	I_C(A)	Test record	R_{th}(K/W)	C_{th}(J/kg • K)
1600	65	0	Fig.6(a)	0.54	3022
1400	85	0	Fig.6(b)	1.17	3085
1600	85	0	Fig.6(c)	1.67	3358
1400	100	0	Fig.6(d)	2.1	3400

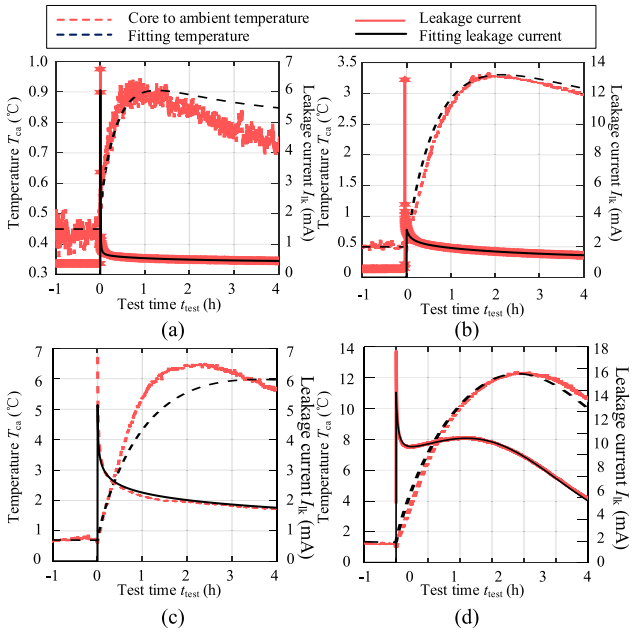


Fig. 6. Thermal response under different testing conditions to obtain resistance and capacitance. (a) 1600 V, 65 °C. (b) 1400 V, 85 °C. (c) 1600 V, 85 °C. (d) 1400 V, 100 °C. The dashed and solid lines denote the fitting curves and measured data for temperature and leakage current, respectively.

can be obtained as follows:

$$R_{\mathrm{ESR}} = R_{\mathrm{s}} + \frac{\tan\delta}{2\pi f C_r}. \tag{3}$$

The thermal resistance from datasheet is tested under certain experimental conditions that are insufficient for precise SOA construction. Thus, experimental characterization should be prioritized for accurate thermal modeling. To obtain the thermal resistance R_{th1} and capacitance C_{th1}, various tests on capacitor 4 have been done, as shown in Table III and Fig. 6.

For DC film capacitors, the axial heat flow path through the end-spray and terminal studs is negligible compared to the radial heat flow path through the dielectric films [11]. Although the intrinsic thermal conductivity of polypropylene is higher in the axial direction, the presence of large contact resistances at the film-to-end-spray interface and the limited effective cross-sectional area of the metallic path result in an axial thermal resistance that is typically one to two orders of magnitude higher than the radial thermal resistance. Radiative heat transfer is also negligible under forced convection conditions with bright metallic surfaces. Thus, based on the heat transfer mechanism illustrated in [12] and considering that the material parameters of

TABLE IV
COMPARISON OF PHYSICAL MODEL AND TEST

V_{DC}	T_a	$R_{th,cal}$	$R_{th,exp}$	$C_{th,cal}$	$C_{th,exp}$
1600	65	0.971	0.54	2957	3022
1400	85	1.217	1.17	3060	3085
1600	85	1.675	1.67	3080	3358
1400	100	2.098	2.1	3196	3400

thermal resistance and heat capacity are temperature-dependent, formula of R_{th1} and C_{th1} are expressed as follows:

$$\begin{aligned} R_{\mathrm{th1}} &= R_{\mathrm{int}}(T) + R_{\mathrm{ext}}(T) \\ &= R_{\mathrm{radial}}(T) + \left(\frac{1}{R_{\mathrm{conv}}(T)} + \frac{1}{R_{\mathrm{cond}}(T)}\right)^{-1} \end{aligned} \tag{4}$$

where, R_{int} is the internal thermal resistance from the heat source to the inner surface of the housing, R_{ext} is the external thermal resistance from the outer surface of the housing to the ambient environment, R_{radial} is the radial thermal resistance from the winding to the inner housing surface through the polypropylene film layers, R_{conv} is the convection thermal resistance from the outer housing surface to ambient air, and R_{cond} is the contact conduction thermal resistance from the housing to the mounting surface

$$C_{\mathrm{th1}} = \sum (m_i \times c_{p,i}(T)) \tag{5}$$

where m_i is the mass of each capacitor material and $c_{\mathrm{p,i}}$ is the corresponding specific heat capacity.

Equations (4) and (5) can be further expressed as (6)–(8) based on [9] and [11]

$$R_{\mathrm{int}} \approx R_{\mathrm{radial}}(T) = \frac{\ln(\frac{D}{d})}{2\pi L \times k_r(T)} \tag{6}$$

$$\begin{aligned} R_{\mathrm{ext}} &= \left(h_{\mathrm{conv}} \times A_{\mathrm{eff}} + h_c(T) \times A_{\mathrm{contact}}\right)^{-1} \\ &= \left(\frac{Nu \times \lambda}{D} \times \left(\pi DL + \pi \frac{D^2}{4}\right) + h_c(T) \times \pi \frac{D^2}{4}\right)^{-1} \end{aligned} \tag{7}$$

where k_{r} is the equivalent radial thermal conductivity, h_{conv} is the convection heat transfer coefficient on the housing surface, A_{eff} is the effective convective surface area of the housing, h_{c} is the contact heat transfer coefficient at the mounting interface, A_{contact} is the contact area between the housing and the mounting surface, *Nu* is Nusselt number, λ is air thermal conductivity

$$\begin{aligned} C_{th1} &= m_{pp} \times c_{pp}(T_0) \times [1 + \alpha_{pp}(T - T_0)] \\ &\quad + m_{PUR} \times c_{PUR}(T_0) \times [1 + \alpha_{PUR}(T - T_0)] \\ &\quad + m_{Al} \times c_{Al}(T_0) \times [1 + \alpha_{Al}(T - T_0)] \end{aligned} \tag{8}$$

where, α_{pp}, α_{PUR}, and α_{Al} are the temperature coefficients of specific heat capacity of polypropylene, polyurethane (PUR) potting resin, and aluminum, T_0 is the reference temperature.

Table IV presents a comparison of R_{th} and C_{th} obtained by experiment and the physical analytical model. The large

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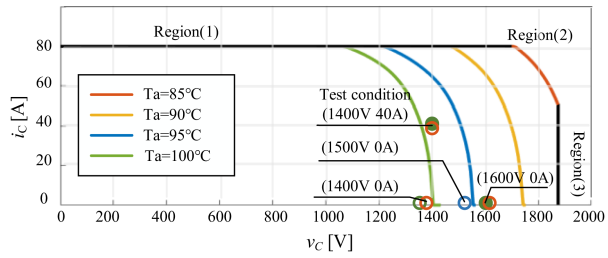


Fig. 7. Improved SOA under different ambient temperature. (Solid circle is broken state and hollow circle is unbroken state.)

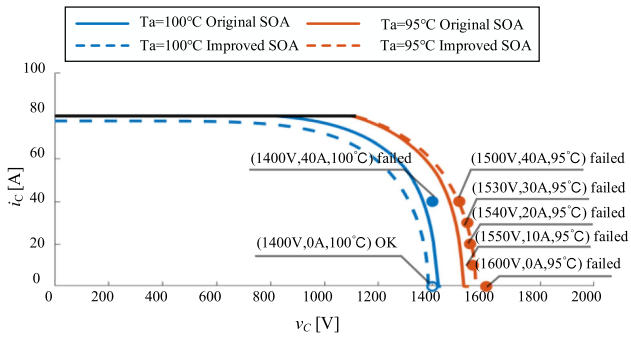


Fig. 8. Comparison of original SOA considering steady-state and improved SOA considering electro-thermal transient under 100°C and 95°C. (Solid circle is broken state and hollow circle is unbroken state.)

deviation at 65 °C is caused by the extremely low temperature rise, which falls within the sensor accuracy, making the identified experimental thermal resistance unreliable. However, for the high-temperature zone (≥85 °C) where capacitors are prone to failure according to the datasheet, the predicted and experimental thermal resistances agree closely, and the thermal capacitances match well across all conditions. Thus, the error in thermal resistance in the low-temperature region does not affect the accurate derivation of SOA.

B. Improved SOA of DC-link Capacitor

Combining the failure mechanism and the transient thermal model, the improved SOA of DC-link capacitor can be determined following the procedure.

1) Obtain the critical positive temperature feedback point of the capacitor and the fixed geometric and material parameters from the datasheet.
2) Determine the exponentially decaying leakage current, DC voltage, and series resistance to calculate the capacitor loss using (2)–(3).
3) Conduct steady-state experimental characterization tests at multiple (T_a, V_{DC}) combinations to calibrate the dataset [temperature coefficients in (6)–(8)].
4) Determine R_{th1} and C_{th1} through iterative electro-thermal coupling analysis for a given operating point (T_a, V_{DC}) by combining (4) and (6)–(8).
5) Calculate the thermal response of the core using (1) to obtain the peak temperature during the transient.
6) Generate the complete SOA boundary curve for a fixed ambient temperature T_a using the bisection method.
7) Obtain SOA based on the positive feedback point and peak current of different conditions, as shown in Fig. 7.

C. Case Studies

In total, seven capacitors same as previous 1.25 kV ones are tested as case studies to verify the proposed SOA modeling method as in Fig. 8. The original SOA is derived from the (3) combined with (5) following [8]. At the ambient temperature at 100 °C, the failed capacitor is outside the SOA. However, the unbroken one which should be within the SOA is out of original SOA. For 95 °C, the broken ones are out of improved SOA but some are within the original SOA. The analysis verifies the necessity of considering the transient electro-thermal process.

In conclusion, Liu et al. [8] obtained the capacitor temperature through the loss during steady-state operation of capacitors and derived the original SOA based on the steady-state temperature. This letter calculates the transient thermal response considering the instantaneous power variation using precise leakage current and temperature-dependent thermal resistance and capacitance. Therefore, the phenomenon observed in [8] where some damaged capacitors were still within steady-state SOA can be attributed partly to parameter tolerances and partly to the use of only steady-state temperature rise without considering the thermal transient. Compared to the original SOA, the improved method can more accurately characterize the SOA region by considering the dynamic thermal processes.

D. Discussion

The primary contribution of this work is a methodology and framework for determining the SOA of DC-link film capacitors considering transient process. Although the framework is initially established under nominal conditions, it can be readily extended to incorporate the effects of operating frequency, thermal environment, and aging as discussed below.

1) High-frequency ripple-induced local hot-spot: At high frequencies, skin effect, proximity effect, and electromagnetic coupling from external connection leads concentrate ripple current at the winding ends near the zinc-spray connections, leading to localized hot spots with temperatures higher than the average core temperature [13]. *tanδ* and electrode loss increase with frequency, so total ESR does not decrease monotonically with *f* but often rises, increasing power dissipation. Consequently, (3) must be used with a frequency-dependent *tanδ*(*f*) and an enhanced $R_s(f)$ to capture this behavior. No modification to DC-related part of (2) is required for the DC voltage boundary as DC leakage loss is frequency independent. T_{core} in (1) should be replaced with local hotspot temperature T_{hot}. To keep T_{hot} below the thermal-runaway limit, I_C must be reduced. Thus, the thermal-runaway temperature boundary shifts inward and the maximum ripple current boundary also moves inward.

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2) Thermal coupling and radiation effects: thermal coupling between adjacent capacitors and mutual radiation primarily affect R_{ext} by reducing A_{eff} in (7) and creating additional heat exchange paths [14]. In the present forced-convection dual-capacitor experiment, the large spacing and identical steady-state temperatures render air conduction and mutual radiation negligible, leaving surface shading as the dominant coupling mechanism. The resulting change in R_{ext} is minor; hence, the electrothermal SOA model itself requires no structural modification.
3) Aging-induced electrothermal parameter degradation: Aging affects the SOA of DC film capacitors mainly through R_{ESR} which increases due to metallization layer corrosion and through insulation resistance R_{p} which increases because of space-charge accumulation and dielectric degradation [15]. Meanwhile, R_{th} and C_{th} remain nearly unchanged, as the polypropylene dielectric does not undergo structural changes below its melting point. Thus, the proposed SOA calculation model is fully applicable without any structural modifications after aging. Based on accelerated aging data, it is only necessary to update the parameters R_{ESR} and I_{lk} in (2), noting that based on basic relation $V_{\mathrm{dc}}/R_{\mathrm{p}}$, the calculation of I_{lk} requires considering a transient peak and an attenuation coefficient.

IV. Conclusion

This letter proposes an improved SOA of the DC-link capacitors based on the catastrophic failure mechanism and transient electro-thermal model. The DC and DC + AC failures of the capacitors are analyzed in detail to determine the common root causes of temperature-induced failure mechanism and its positive temperature feedback process. The transient thermal response is derived based on the loss modeling and the thermal model considering thermal resistance and capacitance under different excitation and temperatures. The SOA is derived based on failure mechanism and transient thermal model. Test results from seven capacitors in case studies reveal that the improved SOA which considers the transient process, it is more accurate than the original SOA which only considers steady-state. The improved SOA is expected to serve as a guideline for sizing of DC-link capacitors.

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